

MPPG 8a

Wednesday, July 31, 2024

11:02 AM

APPENDIX II

MPPG 8.a RECOMMENDED PERFORMANCE TESTS

TABLE I

				Applicable to clinical practice?	QMP initials
Item	Test	Frequency	Tolerance		
Dosimetry tests					
D1	Photon and electron output constancy	Daily ^a	3% of baseline	Y/N	
		Monthly	2% of baseline	Y/N	
		Annual	1% of TG51	Y/N	
D2	Photon and electron beam profile constancy	Daily ^a	2%	Y/N	
		Monthly	2%	Y/N	
		Annual	2% of TPS OAFs ^b	Y/N	
D3	Electron beam energy	Monthly	2 mm	Y/N	
		Annual	2 mm	Y/N	
D4	Photon beam energy	Monthly	1% of PDD/TPR (relative change in value)	Y/N	
		Annual	1% of PDD/TPR at reference depth	Y/N	
D5	Dynamic delivery control	Monthly	3% of open field dose	Y/N	
D6	Photon MU linearity (output constancy)	Annual	2% >10 MU for open field; 2% for segmented field	Y/N	
D7	Electron MU linearity (output constancy)	Annual	2% for clinical range	Y/N	
D8	Photon output vs dose rate	Annual	2%	Y/N	
D9	Photon and electron output vs gantry angle	Annual	2% of IEC gantry 0° output	Y/N	
D10	Photon and electron OAF vs gantry angle	Annual	2% of OAFs at IEC gantry 0°	Y/N	
D11	Arc mode (expected MU, degree)	Annual	2% of MU and 2°	Y/N	
D12	Special procedure mode (TBI/TSET)	Annual	Output: same as regular beam; energy: same as regular beam; profile: same as regular beam	Y/N	
			profile: same as regular beam		
Mechanical tests					
Localizing lasers		Daily	2 mm	Y/N	
		Monthly	1 mm	Y/N	
Optical distance indicator		Daily	2 mm at isocenter	Y/N	
		Monthly	2 mm over clinical range	Y/N	
Jaw position indicators		Daily	2 mm per jaw for single field	Y/N	
		Monthly	2 mm per jaw for clinical range of motion	Y/N	
Light to radiation field coincidence		After Service	2 mm per jaw	Y/N	
Leaf position accuracy		Monthly	1 mm	Y/N	
Gantry/collimator angle indicators		Monthly	1°	Y/N	
Physical graticule (port film graticule)		Monthly	2 mm	Y/N	
Cross-hair centering		Monthly	1 mm	Y/N	

Abs: 2 mm and 1°;

M9	Treatment couch positions (absolute and relative)	Monthly	Rel: 1 mm over 10 cm and 0.5° over 3°	Y/N
M10	Radiation isocentricity (MLC/jaw radiation isocenter with collimator, gantry, and couch rotation)	Annual	2 mm diameter ^c	Y/N
M11	Electron applicator collimator settings/physical inspection/interlocks	Annual	Same as acceptance/TPS	Y/N

TABLE 1

TABLE 1 (Continued)

Item	Test	Frequency	Tolerance	Applicable to clinical practice?	QMP initials
M12	Stereotactic accessories, lockouts, cone coding	Daily Annual	Functional Functional	Y/N Y/N	
M13	Accessory latches/interface (all slots)	Annual	Functional	Y/N	
Safety tests					
S1	Door interlock	Daily	Functional	Y/N	
S2	Door closing safety	After Service	Functional	Y/N	
S3	Audio/visual monitors	Daily	Functional	Y/N	
S4	Beam-on indicator	Daily	Functional	Y/N	
		Annual	Functional (all indicators)	Y/N	
S5	Anticollision test	Daily	Functional (single point for system function)	Y/N	
		Monthly	Functional (all collision interlocks)	Y/N	
S6	Safety procedures	Determined by QMP	Functional	Y/N	
Wedge tests					
W1	Electronic wedge check	Daily	Internal: functional; collimator shaped wedges: 3%	Y/N	
		Monthly	2%	Y/N	
W2	Physical wedge placement accuracy	Monthly	1mm	Y/N	
W3	Wedge profile for 60 degree electronic wedges, all energies	Annual	2% of TPS OAFs	Y/N	
W4	Wedge dose for collimator shaped wedges, all angles	Annual	2% of TPS dose	Y/N	
Comprehensive review of machine settings					
C1	Comprehensive review of machine settings	Annual	Same as acceptance/expected	Y/N	

^a Daily checks should be conducted for the energies used that day.

^b Tolerance is the same as what was achievable for TPS model comparison to measured data as at the time of commissioning.

^c For SRS-SBRT applications, refer to the relevant AAPM Medical Physics Practice Guideline.

Signature of qualified medical physicist:

Date: